

Look What the Probes Dragged In! Investigating Real-World Shortcuts in MedCLIP for Chest X-ray Images

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4th Semester, Spring 2026

Course Code: KISPECI1SE

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Abstract

Vision-language models, such as Contrastive Language-Image Pre-training (CLIP)-based approaches, have reached state-of-the-art (SOTA) results in medical artificial intelligence. However, recent work in this field reveals that CLIP-based models remain vulnerable to shortcuts. We investigate how real-world shortcuts manifest across different layers of the medical CLIP-based model, MedCLIP, and its vision encoder, a frozen ResNet-50. We attach 17 linear classification probes to the intermediate layers of the ResNet-50 and train them on three different dataset configurations: NIH-CXR14 with pneumothorax as its target label and PadChest with both cardiomegaly and pneumothorax as its target labels. This setup allows us to observe model behaviour during evaluation using subgroup-based calibration and layer-wise confidence curves. We find that the final linear probe achieves a high global area under the receiver operating characteristic curve score. Despite this, the curves indicate poor calibration in the models. The layer-wise confidence analyses suggest that shortcuts emerge at different depths. Patterns consistent with localised shortcuts, such as drains, appear at later layers, while patterns consistent with diffuse shortcuts, such as scanner-specific noise patterns, emerge earlier, aligning with previous work. Finally, we conduct a manual analysis of the images, which reveals data quality issues in both NIH-CXR14 and PadChest. Our findings underscore that even SOTA models remain vulnerable to shortcuts, and the need for high-quality and well-annotated datasets to draw solid conclusions. Code can be found on our GitHub¹.

1 Introduction

Medical artificial intelligence (AI) models have shown they can outperform even medical physi-

cians (Shen et al., 2019). Vision-language models (VLMs), a newer architecture that leverages both images and medical reports, have demonstrated high performance results. However, previous work raises questions about the fairness of these models (Luo et al., 2024), and their tendencies to rely on artifacts in the images that make the model perform well on standard benchmarks, but fail on real-world scenarios, namely shortcuts (Geirhos et al., 2020). An example of a shortcut could be a chest drain or a pacemaker, where the model will use this as a decision rule, rather than using clinically relevant features of the image.

Recent work by Sourget et al. (2026a) addresses the lack of robustness to shortcuts in VLMs, and the overall fairness across subgroups, by evaluating six different Contrastive Language-Image Pre-training (CLIP)-based models on chest X-ray datasets. Here, they find that all models show lower performance when evaluated on chest X-ray images without drains than with, indicating the reliance on shortcuts. Further, they find that three of the models, CXR-CLIP, CheXzero and MedCLIP, show evidence of being miscalibrated in a zero-shot classification setting, with predicted probabilities clustered around 0.5 rather than reflecting the actual likelihood of disease. This occurs despite CheXzero and MedCLIP achieving acceptable area under the receiver operating characteristic curve (AUROC) scores.

We further explore these concerns by looking into one of these models, namely MedCLIP (Wang et al., 2022). Previous research shows that models trained on biased data tend to be overconfident (Boland et al., 2025), which makes the miscalibration findings from Sourget et al. worth looking into. If MedCLIP is relying on shortcuts, we expect overconfidence, rather than the probabilities being near random, a pattern Boland et al. examines when investigating the layer-wise confidence using curated biased data. We apply this method-

¹https://github.com/nikodice4/MedCLIP_shortcuts

ology to investigate the MedCLIP model’s layer-wise confidence during evaluation, and examine how real-world shortcuts manifest.

We attach linear classification probes to the intermediate layers of MedCLIP’s pre-trained vision encoder (ResNet-50), as seen in [Figure 1](#). For simplicity, we refer to these probes as layer-wise, although they are attached to individual ResNet-50 building blocks rather than at every layer. After training only the attached probes on three different dataset configurations, we visualise the calibration and confidence curves for each of the 17 linear probes. The three dataset configurations are: pneumothorax classification on NIH-CXR14, cardiomegaly classification on PadChest and pneumothorax classification on PadChest.

Our contributions are therefore as follows:

- Implementing linear classification probes in MedCLIP, and training the probes on real-world medical data, rather than curated biased data.
- Investigating MedCLIP’s vision encoder, ResNet-50, in a linear probing and evaluation setting.
- A manual analysis of the models’ predictions, analysing predictions of high disease confidence, low disease confidence and uncertainty within the chest X-ray datasets.

2 Related Work

A central challenge for medical AI lies in publicly available datasets. [Rafferty and Rajan \(2026\)](#) identify four major dataset challenges: i) label quality and semantic noise introduced by automated annotations, ii) demographic and institutional biases that limit representativeness and create domain shift, iii) technical artefacts and medical devices that promote shortcut learning, and iv) the limitations of evaluation practices. They evaluate four large-scale publicly available datasets: MIMIC-CXR, CheXpert, NIH-CXR14 and PadChest, to evaluate the robustness of different network architectures. They find that no architecture proves to be consistently robust to a domain shift, which emphasises that dataset characteristics are the primary drivers of failure in chest radiography, rather than the network design.

Despite these dataset challenges, deep learning still reaches strong performances on chest X-ray

classification, even without having explicit annotations in the dataset during training. [Tiu et al. \(2022\)](#) find that CheXzero, a CLIP-based model, achieves performance comparable to radiologists. The self-supervised model, CheXzero, is trained without any labelled samples of data, and learns features from raw radiology reports, achieving a mean AUROC of 0.889 on the CheXpert test dataset.

However, high-performance metrics alone do not reveal whether the model is actually learning clinically relevant features; it could instead be a symptom of the model learning shortcuts. A typical shortcut that occurs for the task of pneumothorax classification is when a chest drain is present in a chest X-ray. Chest drains help patients breathe, as they let out air or fluid from the patient’s body, and are also often used in the case of pneumothorax ([McNamara et al., 2026](#)), so the model might exploit this shortcut, if there is a correlation between positive cases and chest drains in the X-rays.

[Oakden-Rayner et al. \(2020\)](#) identify exactly this, where their DenseNet-121 exhibits shortcut learning on the NIH-CXR14 dataset, as it favours images containing a drain over images without a drain for pneumothorax classification. Their AUROC for the chest drains subclass is 0.940, while for images without chest drains, it is 0.770, once again suggesting hidden stratification driven by the drain shortcut.

In line with this research, [Jiménez-Sánchez et al. \(2023\)](#) study two popular datasets, CheXpert and NIH-CXR14, and show that models perform better on the subset containing drains than the one without. When training on approximately 24,000 images, they achieve an AUROC of 0.850 for the drains subset and 0.770 for the subset without drains, which again suggests a reliance on the images with a drain, indicating shortcuts.

These findings of models relying on spurious correlations extend directly to newer architectures such as CLIP-based models. [Sourget et al. \(2026a\)](#) evaluate six CLIP-based models on publicly available chest X-ray datasets and find that they all perform better on X-rays with drains than without, confirming that these newer architectures still lack in robustness, and are not more resilient to shortcuts than standard convolutional neural networks (CNNs). They also find that three of the CLIP-based models investigated produce predicted prob-

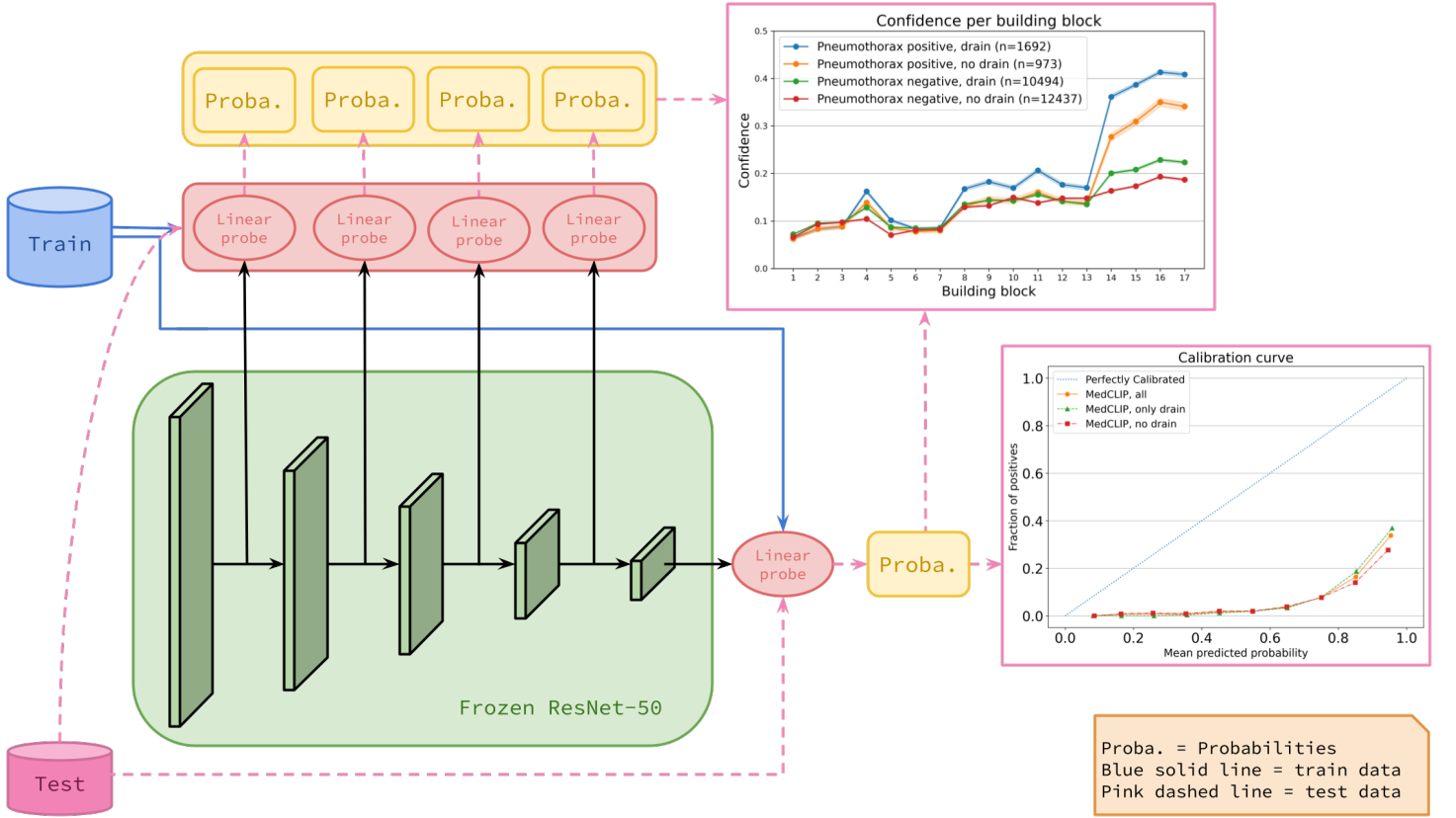


Figure 1: Probe setup for MedCLIP’s frozen ResNet-50 on the NIH-CXR14 pneumothorax classification task. After training the probes, they are evaluated on the test data, and we plot the confidence and calibration curves for each of the defined subgroups. Although ResNet-50 has 50 layers and we train 17 linear probes in total, we show only a representative subset for clarity.

abilities clustered around 0.5 when prompting in a zero-shot setting, even though MedCLIP and CheXzero achieve adequate AUROC scores of 0.88 and 0.80, respectively.

Boland et al. (2025) show that different kinds of shortcuts emerge at different layer depths. They make this observation by training models on data with purposefully added synthetic shortcuts and attaching linear probes to the intermediate layers. In doing so, they visualise how the models’ prediction confidence evolves throughout the layers, when training on biased and unbiased training sets. The synthetic shortcuts used to create the biased training sets consist of diffuse or localised shortcuts. Diffuse shortcuts are uniform random noise in the image, and emerge in the earlier layers, while localised shortcuts, such as a red square placed at a fixed or random location in the images, appear in the later layers. Furthermore, they introduce bias mitigation strategies that target the intermediate layers.

Building on Sourget et al.’s observations, and Boland et al.’s method of using linear probes to investigate shortcuts in the intermediate layers, we aim to identify the depths at which the shortcuts

emerge in a real-world setting. In addition, we do a manual analysis that highlights the need for better data quality in large publicly available medical datasets.

3 Data

We evaluate our linear probes attached to the MedCLIP model, on two publicly available chest X-ray datasets, namely NIH-CXR14 (Wang et al., 2017) and PadChest (Bustos et al., 2020).

The two datasets share similarities, such as the X-rays being taken from PA (postero-anterior) and AP (antero-posterior) projections, but the datasets still differ in their metadata. NIH-CXR14 has simple patient information (gender, age, follow-up) and little information about the image itself, whereas PadChest provides thorough DICOM (Digital Imaging and Communications in Medicine) metadata, including the manufacturer of the scanners, in addition to patient-demographic metadata. Therefore, we found it interesting to investigate both datasets as they have different subgroup characteristics to focus on, such as drains, X-ray machines, and sex. Dataset availability can be found in Appendix A.

3.1 NIH-CXR14

The first dataset, NIH-CXR14, was released in collaboration with the National Institutes of Health (NIH). It is an extension of the NIH-CXR8 dataset, which consists of 108,948 chest X-ray images from 32,717 unique patients. The data contains images collected between 1992-2015 with eight thoracic disease labels (hence the name) mined from the reports using natural language processing techniques. NIH-CXR14 is therefore an extension of the disease labels, adding six more thoracic disease labels to the dataset, and thereby also more images, now containing 112,120 X-rays from 30,805 unique patients. For this dataset, we focus on the disease pneumothorax, also known as collapsed lung. This can occur when air leaks into the space between one’s lung and the chest wall, where the air will push the lung which makes it collapse, either completely or partially (McNamara et al., 2026). We follow their two predefined data split lists: a combined train and validation list and a held-out test list.

Our analysis relies on the chest drains and therefore, we use a supplementary annotation dataset. The first dataset is NEATX (Non-Expert Annotations of Tubes in X-rays) by Cheplygina et al. (2026), who manually annotate chest drains in 3,709 positive samples of pneumothorax from NIH-CXR14 and 1,011 from PadChest, covering different types of tubes. We only use the annotations for NIH-CXR14. As NEATX only contains positive cases of pneumothorax, we use an additional dataset that covers the annotations of negative cases of pneumothorax. Therefore, we use the annotations by Sourget et al., who train a DenseNet-121 on the NEATX labels and apply it to the negative-pneumothorax cases in NIH-CXR14, automatically generating chest drain labels. The drain dataset is therefore a combination of the NEATX dataset by Cheplygina et al. and the automatically generated labels by Sourget et al., giving us the opportunity to analyse drain-related shortcuts in the MedCLIP model.

Our final NIH-CXR14 test set combines the dataset’s original images with chest drain annotations (manually annotated positive-pneumothorax cases) and the automatically generated chest drain annotations from Sourget et al. for the negative-pneumothorax cases.

3.2 PadChest

The second dataset is PadChest (Pathology Detection in Chest radiographs), one of the largest open-source datasets of chest X-ray images, containing multi-labelled annotated reports. It consists of 160,861 images from 67,625 patients collected at the San Juan Hospital of Alicante in Spain between 2009-2017. This also makes it the first dataset to contain Spanish reports of the images.

PadChest covers X-rays from five different projections: PA, L (lateral), AP, AP-horizontal (antero-posterior horizontal) and COSTAL (rib views). We evaluate MedCLIP on two diseases: one with cardiomegaly as the target label, and one with pneumothorax as the target label. Even though the positive class for pneumothorax is heavily underrepresented (0.4%), we still think it is interesting to see the results for the same disease across two different datasets. The reason for choosing cardiomegaly as a target label is its balance between the patient sexes, and its higher class prevalence in the dataset. Cardiomegaly (enlarged heart) can occur from short-term stress on the body, such as a pregnancy, which forces the heart to work harder to ensure enough blood is pumped to the rest of the body (Mayo Clinic Staff, 2022).

It is also worth mentioning that there are three patient sexes in PadChest, “female”, “male” and “other”. We do not exclude “other” from either of the dataset splits, but when we plot the data, “other” is not represented, since only 14 patients fall into this category.

3.3 Preprocessing the Datasets

We process NIH-CXR14 separately for each of its predefined data splits. For the training data, we derive a binary pneumothorax target label based on the “Finding Labels” column. The combined train and validation list is then split into separate training and validation sets, where the validation dataset consists of 20% of the data. We split the training and validation datasets on the patient identifier to ensure no data leakage. For the test dataset, we use the manually and automatically annotated drain dataset, which includes a binary chest drain label and a binary pneumothorax label. For the images, we check for corrupt files, but we do not check for images of just one colour.

For PadChest, we create the binary target labels (cardiomegaly and pneumothorax), and then

Table 1: Complete overview of all datasets, stratified by their splits (training, validation, test) and diagnostic label, with further stratification by the subgroups (sex, scanner, drains) we investigate. The “Total (N)” column reports the total number of images per dataset configuration and split. Each subgroup shows the same images partitioned differently, so the subgroup columns are not additive across categories. Scanner types are abbreviated as IDC (ImagingDynamicsCompanyLtd) and PMS (PhilipsMedicalSystems) and positive and negative are abbreviated as Pos. and Neg.

Dataset	Split	Female Patients		Male Patients		Other Patients		IDC		PMS		Drain		No Drain		Total N	Pos. Ratio (%)
		Neg.	Pos.	Neg.	Pos.	Neg.	Pos.	Neg.	Pos.	Neg.	Pos.	Neg.	Pos.	Neg.	Pos.		
NIH-CXR14 (Pneumothorax)	Train	29,148	1,111	38,314	1,052	-	-	-	-	-	-	-	-	-	-	69,625	3.1
	Val	7,564	243	8,861	231	-	-	-	-	-	-	-	-	-	-	16,899	2.8
	Test	9,483	1,231	13,448	1,434	-	-	-	-	-	-	10,494	1,692	12,437	973	25,596	10.4
	Total	46,195	2,585	60,623	2,717	-	-	-	-	-	-	-	-	-	-	112,120	4.7
PadChest (Cardiomegaly)	Train	30,862	3,518	32,030	2,705	6	1	31,649	3,034	31,249	3,190	-	-	-	-	69,122	9.0
	Val	7,620	881	7,999	689	4	0	7,917	774	7,707	796	-	-	-	-	17,194	9.1
	Test	9,716	1,044	10,183	822	3	0	10,061	909	9,843	957	-	-	-	-	21,770	8.6
	Total	48,198	5,443	50,212	4,216	13	1	49,627	4,717	48,799	4,943	-	-	-	-	108,086	8.9
PadChest (Pneumothorax)	Train	34,287	93	34,580	155	7	0	34,654	29	34,220	219	-	-	-	-	69,122	0.4
	Val	8,483	18	8,653	35	4	0	8,685	6	8,456	47	-	-	-	-	17,194	0.3
	Test	10,734	26	10,949	56	3	0	10,966	4	10,722	78	-	-	-	-	21,770	0.4
	Total	53,504	137	54,182	246	14	0	54,305	39	53,398	344	-	-	-	-	108,086	0.4

we split the data into train, validation, and test sets, splitting on the patient identifier, to ensure no data leakage. 20% of the data goes to the test dataset, then 80% of the data will be training data, and then of those 80%, we take 20% to become the validation data.

For the PadChest datasets, we follow the pre-processing methodology of [Sourget et al. \(2025\)](#), which comprises removing all vertical X-ray projections, keeping only these projections: PA, AP, AP-horizontal. We also exclude rows where the disease label is “suboptimal study”, “exclude” or “Unchanged”. We resize the images to 512×512 pixels and exclude the images that are fully black.

All information about the sizes of the datasets and subgroups can be found in [Table 1](#).

3.4 Errors in the Datasets

The dataset errors for PadChest are described in [Pedersen and Sydendal \(2025\)](#), but for completeness we provide an overview of the dataset errors below:

We reveal that the **PadChest** dataset has registered the sex of some patients incorrectly. We find 12 cases of patients having conflicting sexes in the “PatientSex_DICOM” column. An example of this can be seen for patient identification 1241-7793-5732-0637-9575-1356-9248-1265-9659-936² who has 3 image identifiers and two registered sexes. Investigating further, we see that this patient identification has two distinct birth

²Patient identifiers are formatted with hyphens, for easier readability. The actual patient identifiers in the data have no separators.

years, one image identifier registered as a female born in 1951 and two image identifiers registered as a male born in 1930. This suggests that we are not dealing with a transgender patient, but rather two separate patients who have been incorrectly entered.

Another case we see is for patient identification 1787-9915-0996-8178-8847-5108-3294-3783-4542-059, which has four images registered, but two of those images are identical, even the metadata, with the only differences being the labelling method and the patient sex. The identical images (with differing image identifiers) are registered as both female and male, even though the majority of that patient’s images’ metadata is entered as male. As our focus is not on doing a strict data split on the patient sex to train the model’s linear probes, we do not account for these errors, but we do acknowledge the existence of data quality issues.

Further, for Padchest in [subsection 6.3](#), we reveal that the dataset does not only contain chest X-ray images, as we also find one case of a skull X-ray.

For **NIH-CXR14**, we find 16 cases of patients being older than 100 years. While ages above 100 are possible, the lowest of these entries is 148, which we consider implausible and therefore count this as a data quality issue. During the manual analysis in [subsection 6.3](#), we also encounter a grey image in the data. Once again, we do not exclude the patients with implausible ages or the greyed-out image from the dataset, but we acknowledge that these issues exist.

4 Methodology

4.1 MedCLIP

MedCLIP is a pre-trained VLM combining vision and language encoders. It is pre-trained on CheXpert, MIMIC-CXR, COVID, and RSNA Pneumonia, which consist of chest X-rays and radiology reports for medical image-text representation learning (Wang et al., 2022).

The two main issues that MedCLIP addresses in standard CLIP-based models are that standard CLIP models can only work with limited coupled data, meaning that image and text data need to be paired. However, medical data is already scarce and is therefore not utilised to its full extent. CLIP models also pull semantically paired image and text data (e.g. from the same patient) together in the embedding space, which also results in unpaired image and text data being semantically further apart from each other. Therefore, in the medical setting, this introduces false negatives, since it is not possible to use similar data to describe similar problems that do not originate from the same patient.

MedCLIP, on the other hand, can use unpaired image and text data, thereby also scaling the amount of training data and expanding the multimodality of the model. To address the false negatives, they design a soft semantic matching loss function that measures the semantic similarity between each of the modalities. To understand the setup of the model, we will briefly describe MedCLIP’s architecture and benefits.

MedCLIP is built upon three components:

1. Knowledge extraction of medical entities to build the semantic similarity matrix.
2. Vision and text encoders that extract the image and text embeddings, respectively.
3. A loss function based on the semantic matching loss that trains the whole model.

The **knowledge extraction** module decouples the paired image and text data, which gives more data. If we have n paired image-text data samples, alongside m additional labelled images and h additional medical sentences, the original CLIP method gives us only n paired samples. Using MedCLIP’s decoupling, we get n images and n sentences. When traversing all possible combinations of image-text pairs, we obtain $(n+m) \times (n+h)$,

which results in $\frac{(n+m) \times (n+h)}{n}$ times more data for supervision.

They also leverage external medical knowledge to build the **semantic similarity matrix**. The previous CLIP model treats all positive pairs equally, whereas MedCLIP proposes to instead look at the pairs’ semantic similarity, not just whether the pair is an actual pair or not. To do this, they take the raw medical reports and extract entities using the Unified Medical Language System (UMLS), and map the images’ diagnostic labels to the same UMLS entities. From these entities, they build multi-hot vectors for both the images and text, $\mathbf{l}_{img}$ and $\mathbf{l}_{txt}$. For any image or text instance, it is possible to measure the similarity between them, by comparing $\mathbf{l}_{img}$ and $\mathbf{l}_{txt}$.

The **vision and text encoders** produce the embeddings used to compute the predicted similarity matrix. They encode the X-ray images and medical reports into embeddings, $\mathbf{v} \in \mathbb{R}^D$ and $\mathbf{t} \in \mathbb{R}^M$, respectively, using the vision encoder E_{img} and the text encoder E_{txt} . Two separate projection heads, f_v and f_t , map the raw embeddings down to a shared dimension, making the embeddings directly comparable and suitable for contrastive learning. The dimensions become $\mathbf{v}_p \in \mathbb{R}^P$ and $\mathbf{t}_p \in \mathbb{R}^P$ for the image and text encoders, respectively. This yields

$$\mathbf{v} = E_{img}(\mathbf{x}_{img}) \quad (1a)$$

$$\mathbf{v}_p = f_v(\mathbf{v}) \quad (1b)$$

and

$$\mathbf{t} = E_{txt}(\mathbf{x}_{txt}) \quad (2a)$$

$$\mathbf{t}_p = f_t(\mathbf{t}) \quad (2b)$$

The predicted similarity between an image and text is then the cosine similarity between normalised $\mathbf{v}_p$ and $\mathbf{t}_p$. Computed pairwise, this yields a predicted similarity matrix, $\hat{s}$.

Lastly, the role of the **semantic matching loss function** is to align the predicted similarity matrix from the encoders with the semantic similarity matrix from the UMLS labels, $\mathbf{l}_{img}$ and $\mathbf{l}_{txt}$. During an epoch, they sample N_{batch} input images and texts separately, $\{\mathbf{x}_{image}\}$ and $\{\mathbf{x}_{text}\}$. As mentioned, they do not define positive pairs by searching for equivalent labels, instead they build the semantic similarity matrix s by

$$s = \frac{\mathbf{l}_{img}^\top \cdot \mathbf{l}_{txt}}{\|\mathbf{l}_{img}\| \cdot \|\mathbf{l}_{txt}\|}. \quad (3)$$

The soft targets are obtained by normalising s via softmax in two directions: image-to-text and text-to-image. The cosine similarities between the projected image and text embeddings serve as logits, which are normalised via softmax in combination with the temperature set to 0.07. A temperature smaller than 1 makes the probabilities concentrate on the most likely image-text instance. The semantic matching loss is then the cross entropy between the predicted similarity and the target semantic similarity, computed in both directions and averaged. This averaged value is what the model minimises during training.

4.1.1 ResNet-50 and BioClinicalBERT

In our setup, the text encoder is the BioClinicalBERT (Alsentzer et al., 2019), but we solely use the vision encoder part of MedCLIP. We extract the pre-trained ResNet-50 (He et al., 2016) vision encoder from MedCLIP, instead of the vision transformer (ViT) encoder. The reason we choose ResNet-50 as the vision encoder rather than a ViT is that Boland et al. developed their methods on CNN architectures, and we wish to follow their methodology.

Since we do not train the vision encoder, we freeze all of the parameters of ResNet-50, so we can train the linear probes, ensuring that any signals captured by the probes reflect representations that have already been learned during MedCLIP’s pre-training.

ResNet-50 introduces the deep residual learning framework. Instead of having stacked layers directly fit the desired output, each building block explicitly learns the residual between its input and its output. This is done with skip connections, which have no learnable parameters. It simply adds the block’s input to the output of the stacked layers unchanged. This makes the model learn the residual changes, rather than replicating the input, which makes optimisation easier. ResNet-50 specifically consists of 16 building blocks across four convolutional stages (He et al., 2016).

BioClinicalBERT is based on the original BERT architecture. The weights of the BioClinicalBERT are inherited from BioBERT, which is a further pre-training of the BERT-base on biomedical abstracts and articles from sources such as PubMed. BioClinicalBERT is then subject to continued pre-training on the MIMIC-III (v1.4) clinical note corpora (Alsentzer et al., 2019).

4.2 Linear Probes

To investigate how classification confidence of MedCLIP’s vision encoder develops throughout the layers, we follow the methodology of Boland et al.. Each linear probe consists of an average pooling layer, followed by a single-linear fully connected neural network. They attach the probes to the intermediate layers of their CNN architectures. The linear probes offer insights into a model’s ability to classify layer-wise in a CNN architecture.

We attach linear probes to the output of each building block of our frozen ResNet-50 architecture, and one final probe is placed at the last average pooling layer providing the final predictions for the input, as seen in Figure 1. In total, we train 17 probes. To assess the calibration of the model, we use the predictions from the final probe, the main classification task being disease vs. no disease.

4.3 Training and Evaluating the Probes

We train the attached linear probes three separate times, each time with a different dataset configuration: NIH-CXR14 with pneumothorax as the target label, and PadChest with both cardiomegaly and pneumothorax as the target labels. To train the probes, we use the HPC cluster of the IT University of Copenhagen. We primarily use PURRLab’s partition, an Nvidia A100 40GiB GPU.

In our setup, only the weights of the linear probes are updated, because we use the probes to capture the representations that ResNet-50 learned during MedCLIP’s pre-training. The training loss is the sum of the cross entropy losses across all 17 linear probes (16 are intermediate, plus the final probe on the average-pooled output). Each linear probe has its own cross entropy term in this sum, and since the probes do not share weights and the ResNet-50 is frozen, each probe only receives gradients from its own loss term. This means the gradient of any other loss term with respect to a given probe’s weights is zero, so the other probes’ losses do not affect that probe’s update.

To mitigate overfitting, we check the summed validation loss after each epoch to assess whether the model should do early stopping. We give the model a patience of 15 epochs, so if the model fails to improve within those epochs, training stops. Further, after each epoch, we save the weights in case the model exceeds its time limit on the pro-

vided resources, and to evaluate later on. When evaluating on the held-out test set, the final logits are obtained from the final linear probe, and we apply softmax to obtain the probabilities. Further, we also save the predictions from the intermediate probes attached to the building blocks to construct the confidence curves.

4.3.1 Hyperparameters

Our hyperparameters are inspired by the work of [Cheplygina et al.](#), who perform a grid search over optimal hyperparameters for each of their CNN architectures for both the NIH-CXR14 and PadChest datasets. Even though our probes are single-layer neural networks, we use these hyperparameters because they are data dependent rather than architecture dependent. Since the models do not converge too early or too late (between ~ 40 and ~ 80 epochs) with these hyperparameters, we decide to continue using them. Below is the final hyperparameter list:

- Number of epochs: 100
- Batch size: 32
- Learning rate: 0.00001
- Delta: 0.001
- Patience: 15

4.4 Calibration Curves

We plot the mean predicted probabilities from the final probe against the fraction of true positive cases of the given disease in each bin, to produce our calibration curve. A perfectly calibrated model is one whose predicted probabilities match the true frequency of a given disease. This means, among all cases assigned a probability of p , a fraction p are actually positive. On the plot, this corresponds to the diagonal. If the curve lies below the diagonal, the model is overconfident. Its predicted probabilities then exceed the actual fraction of positives in the bin. If the curve lies above the diagonal, the model is underconfident. Both result in miscalibration.

The purpose of the calibration curve is to ensure correct decision-making, especially in the case of medical AI. When a model is poorly calibrated, it can still achieve high accuracies or AUROC scores but produce probabilities that are either over- or underconfident. An overconfident model can lead

to unnecessary treatments, while an underconfident one can lead to missed cases of disease.

We plot the curves for the full held-out test sets and separately across subgroups: patients with drains and patients without drains for the NIH-CXR14 dataset, and the different scanner manufacturers, ImagingDynamicsCompanyLtd (IDC) and PhilipsMedicalSystems (PMS) for the PadChest dataset, and then patient sexes for both datasets. This is done in order to assess whether the model is systematically biased towards a specific subgroup.

4.5 Confidence Curves

To quantify how classification confidence develops throughout the model, we follow the methodology of [Boland et al.](#), who define the model’s confidence, $C(X)$, as the deviation from the maximum uncertainty of 0.5, where a higher value indicates a greater certainty in the prediction. For a given set of inputs, the confidence formula is defined as:

$$C(X) = \sum_{x \in X} |f(x) - 0.5|, \quad (4)$$

where $f(x)$ is the predicted probability for the positive class. [Boland et al.](#) find that the model becomes more overconfident when training on curated biased data, than when training on clean data. As we are not dealing with synthetically biased data, we do not expect to see the exact same results. Following the taxonomy from [Juodelyte et al. \(2025\)](#), physical devices like chest drains are categorised as external shortcuts. We therefore hypothesise that we will see spikes in the later layers, as the chest drains act more like a localised shortcut than a diffuse one, since a physical device aligns with [Boland et al.](#)’s synthetic red squares rather than the synthetic noise. In contrast, the different X-ray machines are categorised as imaging shortcuts and can produce image noise, which we hypothesise will align with a diffuse shortcut. However, the IDC images also contain an “R” marker in the X-rays, which can resemble a localised shortcut.

4.6 Plot Setup

We evaluate the probes on the different subgroups, based on the given dataset configuration. For the NIH-CXR14 dataset, we evaluate the model’s calibration and confidence on the patients who have and do not have a chest drain in the image. For

both PadChest dataset configurations, with cardiomegaly and pneumothorax as the target labels, we evaluate on the different X-ray machines. For all configurations, we also look into the sexes of the patients.

We also make a bar plot visualising the AUROC scores for each dataset configuration’s subgroup, and their global AUROC scores. We choose AUROC because it measures the model’s ability to discern between positive and negative data instances, meaning it does not set an actual decision threshold. All of the subgroups’ confidence curves are bootstrapped and resampled 1,000 times with the same images at each building blocks’ probe, so we can show the 95% confidence intervals. We do the same for the final probe’s AUROC scores.

5 Results

5.1 Good Overall AUROC Performance Across Dataset Configurations

Figure 2 shows the AUROC scores of the final probe on all the different test set configurations. For pneumothorax classification on NIH-CXR14, the model achieves the lowest global AUROC score across all dataset configurations, with the score at 0.839. For the drain subgroup, we see a small gap of 0.028 between drains (0.840) and no drains (0.812). This gap is even smaller for the subgroup of patient sex, with the female patients achieving a higher AUROC score of 0.840 compared to male patients (0.837).

The PadChest dataset where we classify for cardiomegaly, achieves the highest global AUROC across all three classification tasks, at 0.905. The IDC machine achieves the highest AUROC score across all dataset configurations and subgroups, reaching 0.917, while its counterpart, PMS, reaches 0.893. The female patient subgroup has a higher AUROC score of 0.911, while the male patient subgroup has an AUROC score of 0.897.

Lastly, PadChest with pneumothorax as the target label achieves a global AUROC score of 0.875. PMS and IDC have AUROC scores of 0.801 and 0.852, respectively, with PMS having the lowest AUROC score across all subgroups. We also see the biggest difference between female and male patients here, where female patients reach an AUROC score of 0.895 and male patients reach a score of 0.856.

5.2 Generally Poorly Calibrated Models

Figure 3 shows the calibration curves for the final probe. Each row of subfigures corresponds to one of the three dataset configurations, and all subplots contain three curves split by subgroup. For all configurations, the patient sex subgroup is also plotted.

Figure 3a shows the NIH-CXR14 dataset, with the subgroup representing whether a patient has a drain or not. It is poorly calibrated, as it lies below the diagonal, meaning it shows overconfidence. The green curve, representing images with drains only, is closer to the diagonal, but only marginally when compared to the other curves.

Figure 3c shows the cardiomegaly classification task on PadChest for the different machines used to capture the chest X-rays. These curves are better calibrated, as all three curves are closer to the diagonal, but still show overconfidence. We see no real difference, other than IDC (the green curve) being marginally better calibrated than the two other curves.

Figure 3e shows the pneumothorax classification task on PadChest. This calibration plot shows the most miscalibrated curves, with the highest overconfidence not even reaching a fraction of positive disease instances of 0.20. This time the green curve, which represents the IDC machine, is the worst-calibrated curve not only in this plot, but across all dataset configurations. The PMS curve and the combined curve for both machines are similar in calibration.

Figure 3b, 3d, and 3f all show the patient sex subgroups. These calibration curves all look like their counterparts, Figure 3a, 3c, and 3e, respectively. There is no curve within the sex-related subgroups that does noticeably better than the others.

5.3 Confidence Curves Show Signs of Diffuse and Localised Shortcuts

In Figure 4, each plot has four confidence curves, showing both the subgroups and the positive and negative cases of the given disease.

Figure 4a shows the curves for NIH-CXR14 when classifying for pneumothorax, on the images with and without a drain, where we expect to see higher confidence in the later layers. All four confidence curves remain fairly low and stable across the first 13 building blocks. We see the biggest difference between the groups from building block

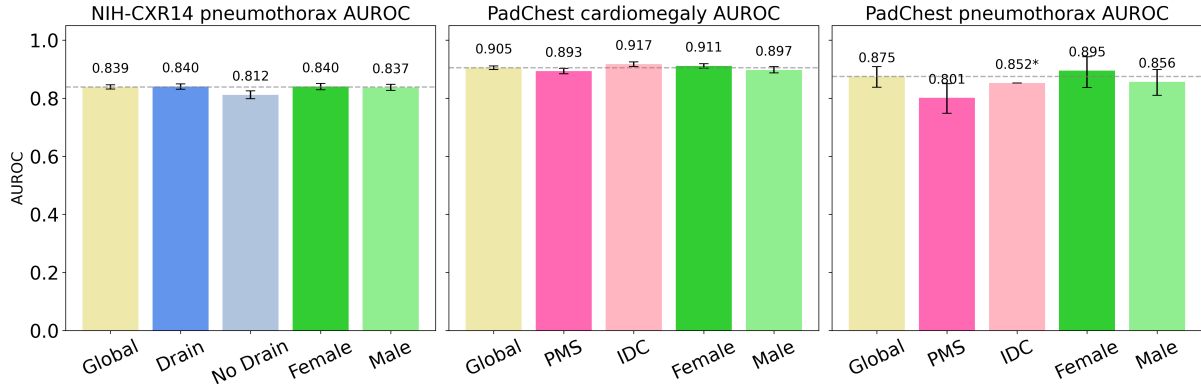


Figure 2: AUROC scores across all dataset configurations and their given subgroups, with the dashed lines indicating the global AUROC scores. The confidence intervals are displayed as black error bars. “*” means that there is no confidence interval due to no variation in the test set.

13, where the positive cases of pneumothorax for both subgroups (drain in blue and no drain in orange) reach high confidence between 0.35 and 0.41. The red and green lines, which are the negative cases of pneumothorax remain relatively low, never reaching a confidence above 0.25, however the green curve (drain) remains higher than the red curve (no drain).

Figure 4c represents the X-ray machines when classifying for cardiomegaly in PadChest. Here we expected to see spikes in the earlier layers. All four curves remain low, with all reaching a spike between building blocks 4 and 6, where the IDC curve (green line) for no cardiomegaly lies above the rest of the curves. Other than that, they remain relatively stable and spike again at building block 13, showing a similar behaviour to Figure 4a. All four lines lie in the same range and yield a relatively low confidence, around 0.23-0.28.

Lastly, for Figure 4e, we see a very different behaviour for PadChest when classifying for pneumothorax, aligning with our expectation of spikes in the earlier layers. Given the only 4 positive pneumothorax cases, the IDC curve (in blue) is very uncertain. All lines spike, once again, at building block 3 and afterwards remain stable in their trajectory. IDC curve (in green) for the negative cases reaches its peak at 0.42, but they all end at similar levels, with PMS for negative pneumothorax cases having the lowest confidence at around 0.19.

Figure 4b, 4d, and 4f look similar to their subgroup counterparts, Figure 4a, 4c, and 4e. The biggest difference appears in the PadChest pneumothorax task (Figure 4f), where the confidence

curves for the negative cases consistently show higher confidence. Both Figure 4e and Figure 4f show lower confidence for the positive cases, but they also have fewer of such cases.

6 Discussion

6.1 Key Takeaways of Results

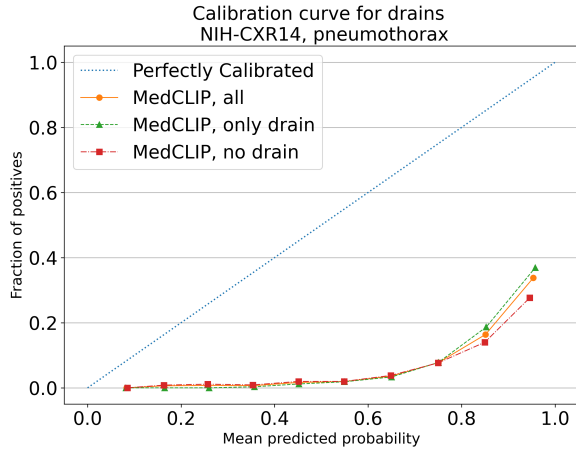
Our results show that the strong AUROC scores in Figure 2 do not translate to the calibration curves. The curves are poorly calibrated, but show a slightly better calibration for the drains subgroup, as seen in Figure 3a.

For the confidence curve in Figure 4a, we see spikes in the later layers, hinting at the model learning shortcuts in favour of the drains. It also shows higher confidence for all of the positive cases of the disease, except for Figure 4e and 4f, which show higher confidence throughout the model for the negative cases, as this is also the dataset configuration with the fewest positive samples.

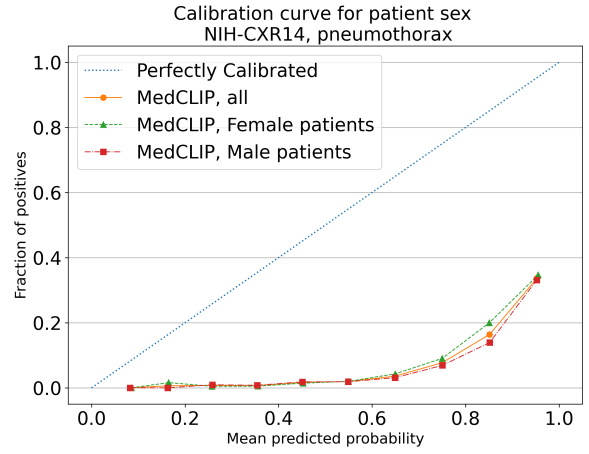
Both the calibration and the confidence curves showed no difference between the patient sexes for any of the dataset configurations.

6.2 Analysis of Results

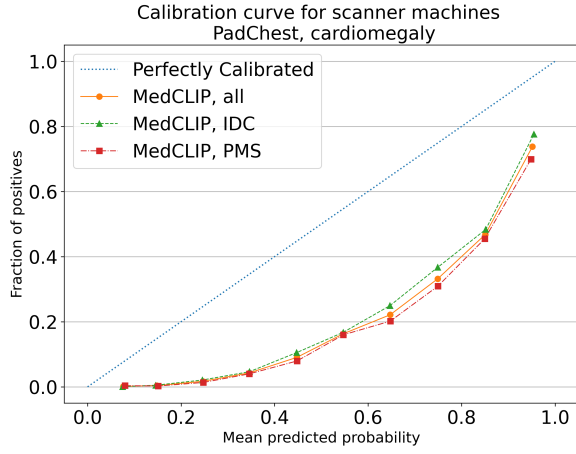
Interpreting our results, we emphasise the importance of evaluating models beyond just performance metrics, such as AUROC. The calibration curves show that even with strong AUROC scores the predicted probabilities can be severely miscalibrated. This is because AUROC is a ranking metric, so the probabilities’ range does not play a role, it depends on their relative ordering in positive-negative pairs.



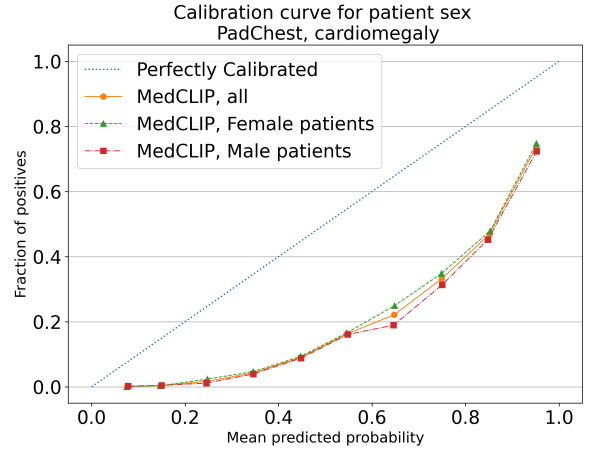
(a) NIH-CXR14 (pneumothorax) calibration curve stratified by drains status.



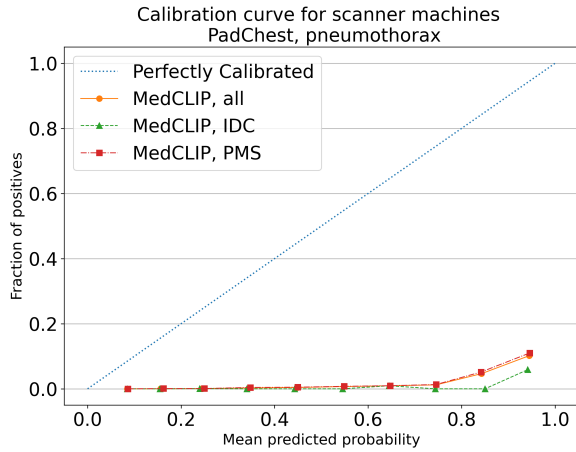
(b) NIH-CXR14 (pneumothorax) calibration curve stratified by patient sex.



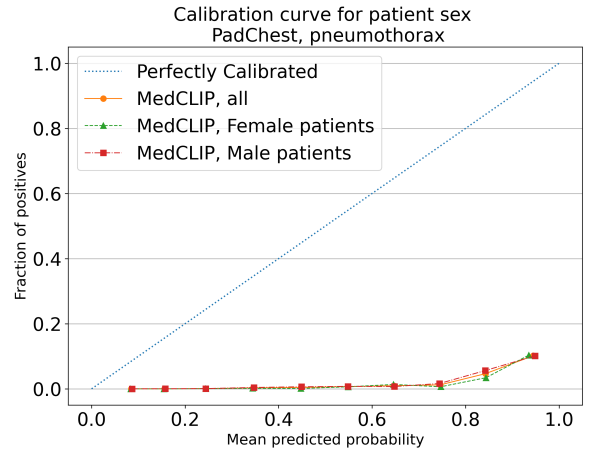
(c) PadChest (cardiomegaly) calibration curve stratified by the X-ray machines.



(d) PadChest (cardiomegaly) calibration curve stratified by patient sex.

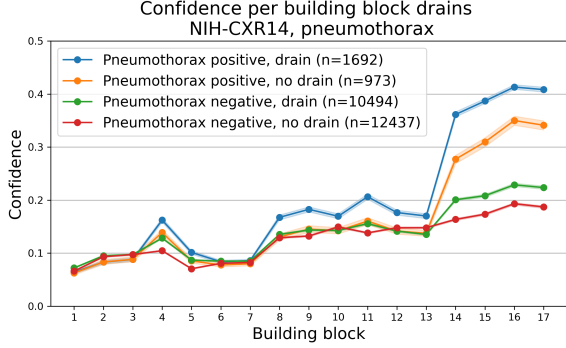


(e) PadChest (pneumothorax) calibration curve stratified by the X-ray machines.

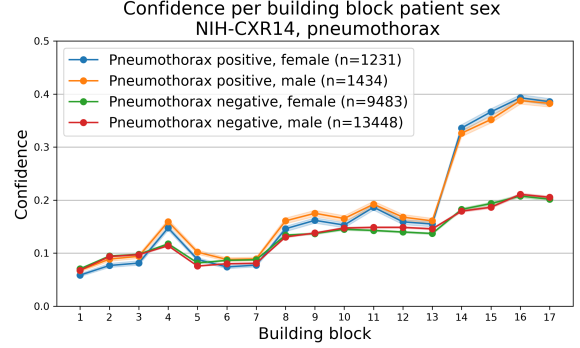


(f) PadChest (pneumothorax) calibration curve stratified by patient sex.

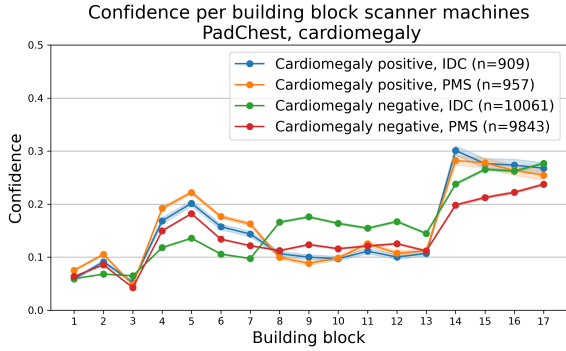
Figure 3: Calibration curves across all dataset configurations and their given subgroups. The x-axis shows the mean predicted probability for each bin, and the y-axis shows the fraction of positive cases of the given disease. The diagonal line represents perfect calibration for the model, while being above or below the line, represent miscalibration.



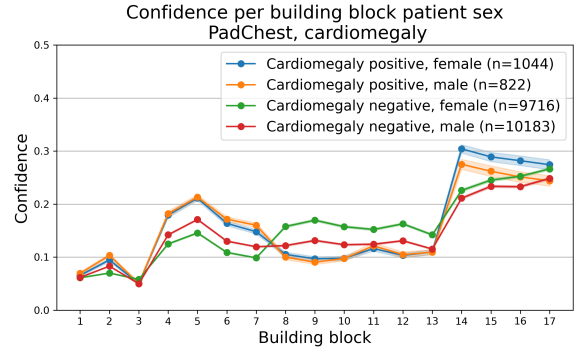
(a) NIH-CXR14 (pneumothorax) confidence curve stratified by drains status.



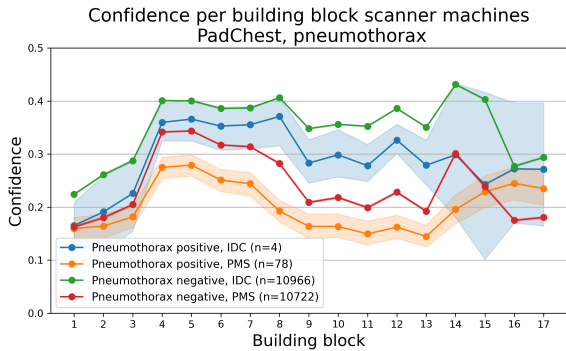
(b) NIH-CXR14 (pneumothorax) confidence curve stratified by patient sex.



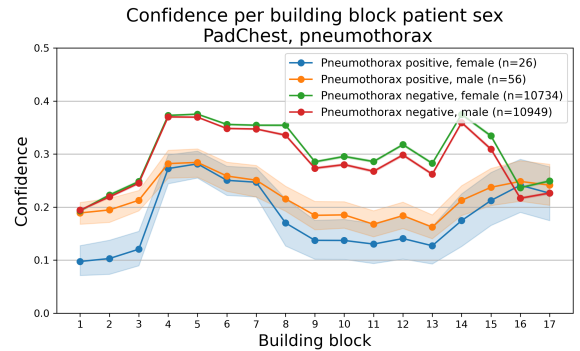
(c) PadChest (cardiomegaly) confidence curve stratified by the X-ray machines.



(d) PadChest (cardiomegaly) confidence curve stratified by patient sex.



(e) PadChest (pneumothorax) confidence curve stratified by the X-ray machines.



(f) PadChest (pneumothorax) confidence curve stratified by patient sex.

Figure 4: Confidence curves across all dataset configurations and their given subgroups. The x-axis represents the individual building blocks that have a probe attached, and the y-axis shows the mean confidence of the model's predictions. 95% confidence intervals for each subgroup's curve are obtained via 1,000 bootstrap resamples per building blocks' probe.

This is again what [Sourget et al.](#) find, with MedCLIP’s probabilities being around 0.5, but still managing to achieve high AUROC scores. Their MedCLIP for NIH-CXR14 pneumothorax classification yielded a global AUROC score of 0.88, while we got 0.839. It is worth mentioning that they do zero-shot classification, while we do linear probing, thereby showing that both classification settings got high AUROC scores, but still show miscalibration. Other CLIP-based models, such as CheXzero, reach AUROC scores in a similar range to ours on PadChest. [Aljaddouh et al. \(2026\)](#) report that CheXzero got an AUROC score (zero-shot classification) of around 0.895 for cardiomegaly and 0.768 for pneumothorax, compared to our global AUROC scores of 0.905 and 0.875, respectively. This once again shows that different CLIP-based models can achieve strong performance, but as our calibration curves demonstrate, this does not guarantee reliable predicted probabilities.

Following [Boland et al.](#) and their definition of localised shortcuts, we see a similar behaviour in our confidence curves for the NIH-CXR14 dataset, with our subgroups of patients having and not having a drain. In [Figure 4a](#) we see a clear spike in the later layers for the positive cases of pneumothorax with drains, which we expect, but the same pattern occurs on the subgroup with no drains (for the positive cases), which is unexpected. Still, the blue curve (drains) has a higher confidence than the orange curve (no drains), and for the negative cases, we see the same pattern. The green curve (drains) has a higher confidence than the red curve (no drains), still leading us to believe that drains are used as a decision rule when predicting. Notably, it is interesting that the orange curve has a confidence of this magnitude. This pattern of medical devices acting as shortcuts is consistent with [Rafferty and Rajan](#)’s third dataset challenge, which is that medical devices can lead models to associate certain devices with diseases, rather than learning clinically relevant features.

We hypothesised a diffuse shortcut could arise for the X-ray machines IDC and PMS, given that imaging shortcuts resemble noise according to the taxonomy from [Juodelyte et al.](#). However, in [Figure 4c](#), the spikes in the earlier layers did not resemble those of [Boland et al.](#), whose curves spike and then follow a stable trajectory, whereas ours decline again. The lack of a shortcut for the X-

ray machines can be due to the fairly balanced data across both machines and their disease status (IDC positive cases: $n = 909$, PMS positive cases: $n = 957$).

For [Figure 4e](#), we do see a pattern of the diffuse shortcut that [Boland et al.](#) see. We see pronounced spikes in the early layers, which overall follow the same trajectory until the last building block. The difference between the machines may lie in the subtle changes when capturing an X-ray, such as different processing algorithms, justifying the pattern of earlier spikes. For PadChest pneumothorax, we also see a big class imbalance between the X-ray machines (IDC positive cases: $n = 4$, PMS positive cases: $n = 78$). These two findings, seeing signs of shortcuts when there is and is not a class imbalance, resemble the findings of [Sourget et al. \(2026b\)](#), who find shortcuts for the different types of X-ray machines, IDC and PMS, when there is a class imbalance, as this is the case for IDC, which is heavily imbalanced. It is important to note that even though we do see signs of diffuse shortcuts in the earlier layers, the IDC X-rays have an “R” marker in the corner, which on its own can resemble a localised shortcut. Therefore, the plots are not enough evidence to conclude whether the shortcuts are diffuse or localised.

Looking at all confidence plots except for PadChest pneumothorax, there seems to be a tendency towards higher confidence for the positive cases, at least in the later layers. This is most noticeable for [Figure 4b](#), where the female and male patients have similar lines, but it also becomes apparent how much more confident the model is at predicting the positive cases than the negative ones.

To summarise, we see how complex models like MedCLIP show clear signs of shortcut learning when we look at the confidence throughout the building blocks. This is especially apparent for the NIH-CXR14 dataset with pneumothorax as the target label. This aligns with the findings of [Sourget et al.](#), however, our observations manifest in the plots made from the linear probes, rather than in the AUROC performance. This suggests that, despite their scale and multi-modality, MedCLIP, and other CLIP-based models are not inherently robust, even though they are state-of-the-art (SOTA) models. Furthermore, with these findings, we show evidence that the synthetic shortcuts presented by [Boland et al.](#) behave in a similar way as real-world shortcuts do in medical datasets.

6.3 Manual Analysis

To get a better understanding of the results, we perform a complementary manual analysis. For each of the three dataset configurations, we look at the final predictions, which come from the last probe. From these predictions, we identify three categories of interest, and within each category, we inspect the top 20 examples in detail and do a visual examination of the top 5 images, plus selected outliers. The full tables are reported in [Appendix B](#). The categories are:

- The examples with the highest predicted probability of disease.
- The examples with the lowest predicted probability of disease.
- The examples the model was the most uncertain about (predicted probability close to 0.5).

NIH-CXR14 pneumothorax: For the images that have a very low probability, three of those images have the annotation “no drain”. However, upon further inspection, all three of those images appear to have a small drain, as we see in [Figure 5a](#), [5b](#), and [5c](#). Also, the target label is “no disease”, meaning this is the data automatically annotated by DenseNet-121. This finding makes us question the quality of the data, and therefore also our drain subgroups, when investigating the shortcuts. The image with the lowest predicted probability, 0.033, is a fully grey image, as seen in [Figure 5d](#), which carries no medical information at all. Upon closer inspection, the pixel values are not perfectly uniform, with values in the greyscale range 61-216. It is surprising that the model is confident in predicting no disease when, in the ideal case, it would simply be uncertain and predict a value around 0.5. Further, for the highest probability images, 19/20 samples have a drain, but only 12 of those are actually true positives. This further provides evidence of MedCLIP, to some degree, relying on the drains when predicting.

PadChest cardiomegaly: Here, there is a pattern of IDC being the most dominant scanner for both the highest and lowest probability images. PMS is more prominent for the uncertain probabilities.

PadChest pneumothorax: We see here that IDC is more present for the lowest probability, and that PMS is more present for the highest and the most uncertain probabilities. Furthermore, when

looking at the actual images for the high and low probabilities, one of the images with a probability of 0.977, taken with the IDC scanner, is a skull X-ray, not a chest X-ray, as seen in [Figure 5e](#).

6.4 Limitations and Ethical Considerations

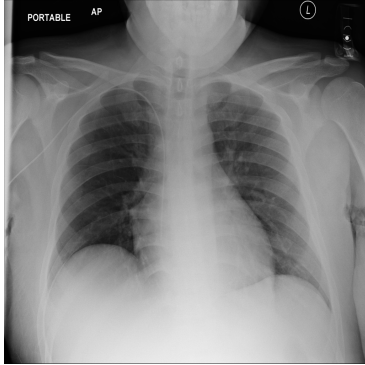
Firstly, we want to note that our hyperparameters are chosen based on the training of CNNs, and not for simple single-layer neural networks. Therefore, some parameters may be too small for the linear probes, such as our learning rate of 0.00001. We do consider this a limitation, but since our convergence was neither too early nor too late, we did not deem this a problem and therefore saw no reason to use resources to re-train the probes.

As for our results, they should be interpreted carefully as we are dealing with two publicly available datasets that contain errors. In NIH-CXR14, we find cases of incorrectly entered patient ages and a greyed-out image. For PadChest, patients have conflicting sexes, duplicate images, duplicate metadata, as well as an X-ray of a skull.

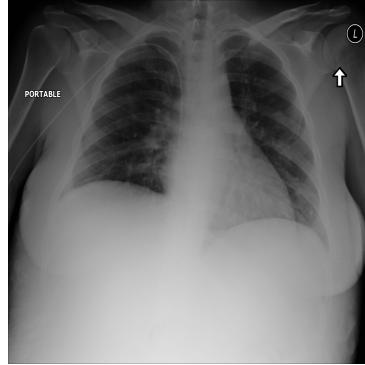
Another limitation in our thesis comes from the automatically annotated drains of the negative cases of pneumothorax in NIH-CXR14. From our manual analysis of the images and samples, we see that even though the labels said “no drain”, upon further inspection, the images do indeed have a drain in them. This does not concern one case, but 3/5 of the images we visually examine. This gives us reason to interpret our results with caution, as the green and red confidence curves in [Figure 4a](#) appear to be stable, but this does not rule out shortcut learning, since the negative pneumothorax annotations are unreliable. This issue of incorrect annotations relates back to [Rafferty and Rajan](#) and their first dataset challenge, which is when annotations are done automatically, it can introduce noise. Therefore, we suggest that either experts or non-experts look through a sample of the negative-pneumothorax images and manually annotate them.

These errors in both datasets make us question the validity of our experiments given the quality of the datasets, as now most of the conclusions we draw should be second-guessed.

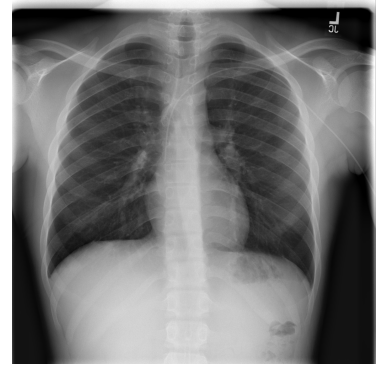
It is also apparent how methods such as calibration curves and confidence plots can be severely influenced by the number of positive cases available, which we see for PadChest with pneumothorax as the target label.



(a) NIH-CXR14 (image ID 00020945_039.png) has no pneumothorax. A small drain can be seen in the top left corner, but it was annotated as having no drain.



(b) NIH-CXR14 (image ID 00013377_010.png) has no pneumothorax. A small drain can be seen in the top left corner, but it was annotated as having no drain.



(c) NIH-CXR14 (image ID 00017318_015.png) has no pneumothorax. A small drain can be seen in the top right corner, but it was annotated as having no drain.



(d) NIH-CXR14 (image ID 00010007_121.png) is a non-uniform grey image.



(e) PadChest (image ID 216840111366964012339356563862009068132653048_00-050-189.png) is not a chest X-ray, but an X-ray of a skull, annotated with lung diseases.

Figure 5: Selected standout images identified during the manual analysis.

For PadChest, we see cases of patients having their sex entered as “other”, which we have not accounted for. These 14 patients are included when training, but they are not represented in the patient sex subgroups, which is something to keep in mind. The NIH-CXR14 dataset patient sex column is also called “Patient Gender”, but the column itself has values called “F” and “M”. We deem this an error in the dataset, as gender does not equal sex, and the use of “female” and “male” typically refers to biological sex rather than gender identity, which the column name suggests (Council of Europe, 2019).

6.5 Future Work

For future research, it would be worth investigating the X-ray view projection as a subgroup. In the PadChest dataset, cardiomegaly has a high class imbalance between the projections, PA, AP and AP-horizontal, sample counts per projection can be found [Appendix C](#). These projections are taken from different positions: in PA projection, the patient is standing up with their back to the X-ray source, and their chest closest to the detector, while in AP projections the patient is sitting down facing the X-ray source, with their chest farthest away from the detector. PA projections are of higher quality and can assess heart sizes better, so radiologists recommend that patients have X-rays taken in the PA position. However, this is not always possible, as patients are required to stand up for those X-rays, which is dependent on the patient’s health (Lloyd-Jones, 2019).

This imbalance in the cardiomegaly class, can emerge naturally as PA is the preferred projection for assessing heart size. In AP the heart appears magnified, since it sits farther from the detector, which can make normal hearts look enlarged. This makes for a typical scenario for a shortcut to arise. The model could rely on projection-specific features, rather than on clinically relevant features for cardiomegaly.

7 Conclusion

In this paper we applied the methodology of linear probes in the intermediate layers from [Boland et al.](#) to a new model architecture, namely MedCLIP. We studied how real-world shortcuts emerge in the intermediate layers by attaching linear probes to three separate MedCLIP models’ vision encoders, particularly ResNet-50’s building

blocks, across three different dataset configurations: NIH-CXR14 for pneumothorax, and PadChest for cardiomegaly and pneumothorax.

Our final linear probes’ predictions achieved acceptable global AUROC scores (0.839, 0.905, 0.875), with only small gaps between the subgroups. However, these AUROC scores did not translate to our calibration curves, given the ranking behaviour of the AUROC metric. Across all three dataset configurations, all calibration curves were positioned below the diagonal, meaning all curves were overconfident in some way, with PadChest pneumothorax being severely miscalibrated.

The confidence curves further revealed that the confidence did not develop in a stable manner across the network, but aligned with [Boland et al.](#), who state that different shortcuts manifest at different depths of the network. This finding contributes to the fact that SOTA models like MedCLIP, regardless of the complexity or scale, are not inherently robust to shortcuts.

Our manual analysis revealed unreliable drain annotations for the negative-pneumothorax samples in NIH-CXR14, emphasising how incorrect automatic labelling can introduce noise. For this reason, we cannot draw thorough conclusions, given the dataset foundations, both in terms of the unreliable automatic annotations and the poor data quality of the two publicly available chest X-ray datasets, NIH-CXR14 and PadChest, given the errors we found.

Overall, our findings demonstrate that a high AUROC score does not indicate whether the model is well-calibrated. We further show evidence that real-world shortcuts, such as drains, manifest in similar ways to synthetic shortcuts. The manual analysis underscores the importance of ensuring good data quality, both in the metadata and when dealing with automated labelling, so it is possible to provide reliable conclusions about SOTA models.

References

- Batoul Aljaddouh, D. Malathi, and Feisal Alaswad. 2026. Multimodal vision-language models in chest x-ray analysis: a study of generalization, supervision, and robustness. In *Biomedical Engineering Letters*, page 517–537, Cham. Springer Nature Switzerland.
- Emily Alsentzer, John Murphy, William Boag, Wei-Hung Weng, Di Jindi, Tristan Naumann, and

- Matthew McDermott. 2019. [Publicly available clinical BERT embeddings](#). In *Proceedings of the 2nd Clinical Natural Language Processing Workshop*, pages 72–78, Minneapolis, Minnesota, USA. Association for Computational Linguistics.
- Christopher Boland, Sotirios A. Tsaftaris, and Sonia Dahdouh. 2025. [Preventing shortcut learning in medical image analysis through intermediate layer knowledge distillation from specialist teachers](#). *Machine Learning for Biomedical Imaging*, 3:447–476.
- Aurelia Bustos, Antonio Pertusa, Jose-Maria Salinas, and Maria de la Iglesia-Vayá. 2020. [PadChest: A large chest x-ray image dataset with multi-label annotated reports](#). 66:101797.
- Veronika Cheplygina, Cathrine Damgaard, Trine Naja Eriksen, Dovile Juodelyte, and Amelia Jiménez-Sánchez. 2026. Augmenting chest x-ray datasets with non-expert annotations. In *Medical Image Understanding and Analysis*, pages 133–144, Cham. Springer Nature Switzerland.
- Council of Europe. 2019. [Gender matters](#).
- Robert Geirhos, Jörn-Henrik Jacobsen, Claudio Michaelis, Richard Zemel, Wieland Brendel, Matthias Bethge, and Felix A. Wichmann. 2020. [Shortcut learning in deep neural networks](#). 2(11):665–673.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. 2016. [Deep residual learning for image recognition](#). In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778.
- Amelia Jiménez-Sánchez, Dovile Juodelyte, Bethany Chamberlain, and Veronika Cheplygina. 2023. [Detecting shortcuts in medical images - a case study in chest x-rays](#). In *2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI)*, pages 1–5.
- Dovile Juodelyte, Yucheng Lu, Amelia Jiménez-Sánchez, Sabrina Bottazzi, Enzo Ferrante, and Veronika Cheplygina. 2025. Source matters: Source dataset impact on model robustness in medical imaging. In *Applications of Medical Artificial Intelligence*, pages 105–115, Cham. Springer Nature Switzerland.
- Graham Lloyd-Jones. 2019. [Chest x-ray quality](#).
- Yan Luo, Min Shi, Muhammad Osama Khan, Muhammad Muneeb Afzal, Hao Huang, Shuaihang Yuan, Yu Tian, Luo Song, Ava Kouhana, Tobias Elze, Yi Fang, and Mengyu Wang. 2024. Fairclip: Harnessing fairness in vision-language learning. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12289–12301.
- Mayo Clinic Staff. 2022. [Enlarged heart](#).
- Caoimhe McNamara, Meadhbh Ní Fhlatharta, and Donna A Eaton. 2026. [Pneumothorax and chest drain insertion](#). *Surgery (Oxford)*, 44(3):197–202. Cardiothoracic Surgery II.
- Luke Oakden-Rayner, Jared Dunnmon, Gustavo Carneiro, and Christopher Ré. 2020. [Hidden stratification causes clinically meaningful failures in machine learning for medical imaging](#). In *Proceedings of the ACM Conference on Health, Inference, and Learning*, pages 151–159. Association for Computing Machinery.
- Nikollette Pedersen and Regitze Sydendal. 2025. Impact of demographic data diversity on model testing dynamics using data cartographies. Msc research project, IT University of Copenhagen, December. Course code: KIREPRO1PE. Supervisors: Théo Sourget and Veronika Cheplygina.
- Amy Rafferty and Ajitha Rajan. 2026. [Limitations of public chest radiography datasets for artificial intelligence: Label quality, domain shift, bias and evaluation challenges](#).
- Jiayi Shen, Casper J.P. Zhang, Bangsheng Jiang, Jiebin Chen, Jian Song, Zherui Liu, Zonglin He, Sum Yi Wong, Po-Han Fang, and Wai-Kit Ming. 2019. [Artificial intelligence versus clinicians in disease diagnosis: Systematic review](#). *JMIR Medical Informatics*, 7(3):e10010.
- Théo Sourget, David Restrepo, Céline Hudelot, Enzo Ferrante, Stergios Christodoulidis, and Maria Vakalopoulou. 2026a. Fairness and robustness of clip-based models for chest x-rays. In *Fairness of AI in Medical Imaging*, pages 11–21, Cham. Springer Nature Switzerland.
- Théo Sourget, Niclas Claßen, Jack Junchi Xu, Rob van der Goot, and Veronika Cheplygina. 2026b. [Dataset diversity metrics and impact on classification models](#).
- Théo Sourget, Michelle Hestbek-Møller, Amelia Jiménez-Sánchez, Jack Junchi Xu, and Veronika Cheplygina. 2025. [Mask of truth: Model sensitivity to unexpected regions of medical images](#). *Journal of Imaging Informatics in Medicine*.
- Ekin Tiu, Ellie Talus, Pujan Patel, Curtis P. Langlotz, Andrew Y. Ng, and Pranav Rajpurkar. 2022. [Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning](#). *Nature Biomedical Engineering*, 6(12):1399–1406.
- Xiaosong Wang, Yifan Peng, Le Lu, Zhiyong Lu, Mohammadhadi Bagheri, and Ronald M. Summers. 2017. [Chestx-ray8: Hospital-scale chest x-ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases](#). In *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, page 3462–3471. IEEE.

Zifeng Wang, Zhenbang Wu, Dinesh Agarwal, and Jimeng Sun. 2022. [MedCLIP: Contrastive learning from unpaired medical images and text](#). In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 3876–3887, Abu Dhabi, United Arab Emirates. Association for Computational Linguistics.

Appendix

A Data Availability

All datasets used are publicly available:

- NIH-CXR14: <https://www.kaggle.com/datasets/nih-chest-xrays/data>
- PadChest: <http://bimcv.cipf.es/bimcv-projects/padchest/>
- NEATX: <https://zenodo.org/records/14944064>

B Manual Analysis Across Dataset Subgroups

Table 2: Manual analysis of the final probe’s predictions on the NIH-CXR14 test set for pneumothorax classification. The table shows the top 20 images for three categories: most confident in predicting no pneumothorax (probability closest to 0), most confident in predicting pneumothorax (probability closest to 1), and most uncertain (probability closest to 0.5). Rows where the drain subgroup is in the minority for that category are underlined. Having a drain is 1 and not having a drain is 0. Probabilities are rounded to 5 decimals.

Image Identifier	True Label	Drain Status	Patient Sex	Probability
Most confident in predicting no pneumothorax for NIH-CXR14 (probability closest to 0)				
00010007_121.png	0	1	M	0.03302
00020945_039.png	0	<u>0</u>	M	0.07243
00030450_001.png	0	1	M	0.07381
00013377_010.png	0	<u>0</u>	F	0.07439
00030450_002.png	0	1	M	0.08059
00017318_015.png	0	<u>0</u>	M	0.08154
00013256_005.png	0	<u>0</u>	F	0.08522
00017504_010.png	0	1	F	0.08524
00023176_019.png	0	1	M	0.08804
00016184_004.png	0	1	F	0.09618
00001247_006.png	0	<u>0</u>	M	0.09799
00014996_005.png	0	<u>0</u>	M	0.09874
00014996_008.png	0	1	M	0.09890
00030522_001.png	0	1	M	0.09998
00020482_003.png	0	<u>0</u>	F	0.10307
00015530_156.png	0	<u>0</u>	M	0.10419
00021420_004.png	0	<u>0</u>	M	0.10625
00014798_000.png	0	<u>0</u>	M	0.10643
00029715_002.png	0	1	M	0.10679
00026412_006.png	0	1	M	0.11061
Most confident in predicting pneumothorax for NIH-CXR14 (probability closest to 1)				
00016568_023.png	1	1	F	0.99956
00013111_057.png	0	1	M	0.99955
00022707_012.png	0	1	F	0.99950
00022707_011.png	0	1	F	0.99948
00022726_006.png	1	1	F	0.99947
00016568_037.png	1	1	F	0.99924
00015078_010.png	1	1	M	0.99919
00007670_005.png	1	1	F	0.99916
00022726_008.png	1	1	F	0.99916
00010936_018.png	1	1	F	0.99916
00016568_036.png	1	1	F	0.99915
00016587_085.png	1	1	F	0.99906
00008841_055.png	0	1	F	0.99903

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Image Identifier	True Label	Drain Status	Patient Sex	Probability
00022726_007.png	0	1	F	0.99903
00012622_031.png	0	1	F	0.99886
00010936_016.png	1	1	F	0.99885
00022726_005.png	0	1	F	0.99883
00009349_025.png	1	1	M	0.99882
00019124_091.png	1	1	F	0.99882
00027478_004.png	0	<u>0</u>	M	0.99878
Most uncertain in predicting pneumothorax for NIH-CXR14 (probability closest to 0.5)				
00018387_027.png	0	1	M	0.50000
00000830_000.png	0	<u>0</u>	F	0.50001
00007557_017.png	0	1	M	0.49999
00011514_012.png	0	1	M	0.50003
00004822_012.png	0	1	M	0.50003
00003528_038.png	0	<u>0</u>	M	0.50003
00012364_008.png	0	1	F	0.50004
00000181_051.png	0	1	F	0.50004
00015605_046.png	0	<u>0</u>	F	0.50006
00016522_003.png	0	<u>0</u>	M	0.49994
00029879_001.png	0	<u>0</u>	M	0.49992
00021201_103.png	0	1	M	0.50008
00013993_034.png	0	1	M	0.49992
00021796_008.png	0	1	M	0.49986
00014798_016.png	0	<u>0</u>	M	0.49985
00020408_001.png	0	<u>0</u>	M	0.50016
00000506_020.png	0	<u>0</u>	M	0.50016
00013249_046.png	0	1	M	0.50017
00011579_040.png	0	<u>0</u>	M	0.49982
00019495_007.png	0	<u>0</u>	M	0.50019

Table 4: Manual analysis of the final probe's predictions on the PadChest test set for cardiomegaly classification. The table shows the top 20 images for three categories: most confident in predicting no cardiomegaly (probability closest to 0), most confident in predicting cardiomegaly (probability closest to 1), and most uncertain (probability closest to 0.5). Rows where the scanner subgroup is in the minority for that category are underlined. Scanner types are abbreviated as IDC (ImagingDynamicsCompanyLtd) and PMS (PhilipsMedicalSystems). Probabilities are rounded to 5 decimals.

Image Identifier	True Label	Scanner Type	Patient Sex	Probability
Most confident in predicting no cardiomegaly for PadChest (probability closest to 0)				
216840111366964013686042548532013156114347330_02-105-039.png	0	IDC	F	0.02615
103693917093872830847809526276065588596_q5hqnk.png	0	IDC	F	0.02623
216840111366964012959786098432011039154446686_00-178-047.png	0	IDC	F	0.02825
216840111366964012487858717522009210154307867_00-030-109.png	0	<u>PMS</u>	F	0.02854
216840111366964013515091760022012320082235519_01-151-073.png	0	IDC	F	0.02898
216840111366964012734950068292010145101156547_03-149-111.png	0	IDC	F	0.02977
216840111366964012558082906712009328132833379_00-102-030.png	0	IDC	F	0.02980
216840111366964012819207061112010302091108120_03-189-037.png	0	<u>PMS</u>	M	0.02999
216840111366964012283393834152009015142006557_00-010-104.png	0	IDC	M	0.03185
216840111366964013076187734852011214114344155_00-117-165.png	0	IDC	F	0.03210
216840111366964012373310883942009198123827814_00-028-192.png	0	IDC	F	0.03297
216840111366964012904401302362010327130256981_04-020-169.png	0	IDC	F	0.03297
216840111366964012959786098432011046115852691_00-175-070.png	0	IDC	F	0.03359
216840111366964013370802480092012137101014784_01-076-055.png	0	IDC	M	0.03365
216840111366964012768025509942010189082006475_03-134-044.png	0	IDC	F	0.03387
216840111366964013590140476722013036105134441_02-059-101.png	0	IDC	F	0.03409
216840111366964013686042548532013233123120764_02-004-096.png	0	IDC	M	0.03411
216840111366964012989926673512011123121200425_00-152-069.png	0	IDC	F	0.03511
216840111366964012959786098432011048120334955_00-174-198.png	0	IDC	F	0.03514
216840111366964013402131755672012186101415753_01-066-065.png	0	IDC	M	0.03518
Most confident in predicting cardiomegaly for PadChest (probability closest to 1)				
103953050703210888484377088771966492555_pnj9nk.png	1	IDC	F	0.99915
216840111366964013076187734852011293122719023_00-195-156.png	1	<u>PMS</u>	M	0.99915
216840111366964012922382741642010356134221543_04-000-079.png	1	IDC	F	0.99909
216840111366964013340662495472012102131826972_01-073-154.png	1	IDC	F	0.99887
216840111366964012373310883942009159091117491_00-092-146.png	1	IDC	F	0.99886
216840111366964013217898866992012005094601199_01-041-018.png	1	IDC	F	0.99876
310081740271657206149132649249885443756_21ruxn.png	1	<u>PMS</u>	F	0.99851
216840111366964013686042548532013142110140059_02-042-097.png	1	IDC	M	0.99845
216840111366964012734950068292010140095847842_03-152-154.png	1	IDC	M	0.99830
216840111366964012989926673512011062131544336_00-163-185.png	1	IDC	M	0.99825
88384111141018197482738265860146005752_hpuu0z.png	0	<u>PMS</u>	F	0.99814
216840111366964013340662495472012115103657225_01-108-199.png	1	IDC	M	0.99794
216840111366964012283393834152009016133737957_00-007-031.png	1	IDC	F	0.99761
216840111366964012959786098432011059084950483_00-179-199.png	1	IDC	F	0.99725
319140003536142580429926023502032037994_c6m6hy.png	1	<u>PMS</u>	M	0.99700
266463130540256183061556297994476560538_jkbs4h.png	1	<u>PMS</u>	F	0.99682
216840111366964012904401302362010330090715155_04-023-175.png	1	IDC	M	0.99674
147894182338319973517835584810509013488_df5svm.png	1	IDC	F	0.99672
216840111366964012558082906712010040084044056_03-166-167.png	1	IDC	F	0.99668
216840111366964013829543166512013339121511185_02-090-031.png	0	IDC	M	0.99661
Most uncertain in predicting cardiomegaly for PadChest (probability closest to 0.5)				
216840111366964013076187734852011173090611529_01-019-081.png	0	<u>PMS</u>	M	0.50000
216840111366964013076187734852011264140425121_00-139-052.png	0	<u>PMS</u>	F	0.50006
216840111366964012558082906712010083112249237_03-153-192.png	0	<u>PMS</u>	M	0.49992
89532200209511112896356230533830256387_2_m7y0a5.png	0	<u>PMS</u>	M	0.50021
216840111366964012558082906712009323115604963_00-107-023.png	1	<u>PMS</u>	F	0.49976
43315893044663659730502047730686707718_542wx6.png	0	<u>PMS</u>	F	0.50024
254061600326412664608949562576866968807_s7siai.png	0	<u>PMS</u>	F	0.49975
216840111366964012768025509942010215143906410_02-120-135.png	0	<u>PMS</u>	M	0.50026
216840111366964012373310883942009119095142241_00-068-036.png	0	<u>IDC</u>	M	0.50031
322104729448612865477048880977107099976_2_711063.png	0	<u>IDC</u>	F	0.50035
216840111366964012768025509942010180143017585_04-008-163.png	1	<u>PMS</u>	M	0.50038
220548764618816410916687487931024157980_thex64.png	0	<u>IDC</u>	M	0.49958
216840111366964012768025509942010204105324219_03-139-118.png	0	<u>PMS</u>	F	0.50046

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Image Identifier	True Label	Scanner Type	Patient Sex	Probability
216840111366964012819207061112010264131648049_03-130-117.png	0	IDC	M	0.50054
216840111366964012989926673512011151135509306_01-014-089.png	0	PMS	F	0.50056
216840111366964012558082906712009309090123748_00-103-059.png	0	PMS	M	0.49940
216840111366964012959786098432011055095657054_00-174-073.png	0	IDC	F	0.49937
216840111366964012373310883942009176082010724_00-095-074.png	0	IDC	F	0.49931
111672055128076644165465972262483677901_bpnt0i.png	0	PMS	M	0.49929
216840111366964013829543166512013337111451074_02-090-165.png	0	IDC	F	0.50076

Table 6: Manual analysis of the final probe's predictions on the PadChest test set for pneumothorax classification. The table shows the top 20 images for three categories: most confident in predicting no pneumothorax (probability closest to 0), most confident in predicting pneumothorax (probability closest to 1), and most uncertain (probability closest to 0.5). Rows where the scanner subgroup is in the minority for that category are underlined. Scanner types are abbreviated as IDC (ImagingDynamicsCompanyLtd) and PMS (PhilipsMedicalSystems). Probabilities are rounded to 5 decimals.

Image Identifier	True Label	Scanner Type	Patient Sex	Probability
Most confident in predicting no pneumothorax for PadChest (probability closest to 0)				
216840111366964012558082906712009322124844600_00-107-093.png	0	IDC	F	0.04158
216840111366964012558082906712009344110636318_00-089-099.png	0	IDC	F	0.04271
248261555152019367507591328512148732003_yot61b.png	0	IDC	M	0.04431
147036015370488755117195440122198442935_gj6nyv.png	0	IDC	F	0.04608
208990610239313623398510691329973957149_ttvss6.png	0	IDC	M	0.04898
216840111366964013217898866992012020113614271_01-043-003.png	0	IDC	F	0.04933
216840111366964012558082906712009321142754309_00-107-079.png	0	IDC	M	0.04963
216840111366964013590140476722013073112859319_02-050-080.png	0	IDC	F	0.05022
216840111366964013663026955732013109102420404_02-074-091.png	0	<u>PMS</u>	F	0.05129
216840111366964012339356563862009047135330129_00-060-137.png	0	IDC	F	0.05135
216840111366964012131105513272008164125919600_00-093-087.png	0	IDC	F	0.05162
216840111366964012487858717522009240094503194_00-005-167.png	0	IDC	M	0.05249
214468601997567053028776045301281842609_nlykuo.png	0	IDC	F	0.05295
284399022025784412265141793198645270023_hl1810.png	0	IDC	F	0.05310
216840111366964012558082906712010033134024938_04-009-115.png	0	IDC	F	0.05369
160270690107571403147404683601658494332_q07f9u.png	0	IDC	F	0.05429
1275224347932024851065_00-041-170.png	0	IDC	M	0.05430
216840111366964013686042548532013157094834755_02-103-033.png	0	IDC	F	0.05528
234532423239700766203011134203007488730_4mefoi.png	0	IDC	M	0.05630
216840111366964012819207061112010294095632878_03-193-097.png	0	IDC	F	0.05679
Most confident in predicting pneumothorax for PadChest (probability closest to 1)				
216840111366964013590140476722013081090219068_02-051-197.png	0	PMS	M	0.99991
216840111366964013590140476722013073124742612_02-055-150.png	0	PMS	M	0.99986
235635165976594378061468822488815969962_5aqavg.png	0	PMS	M	0.99943
216840111366964013590140476722013071210407803_02-053-116.png	0	PMS	M	0.99918
216840111366964012558082906712010081091913043_03-161-159.png	1	PMS	M	0.99820
216840111366964013590140476722013079143704872_02-050-119.png	0	<u>IDC</u>	M	0.99772
216840111366964013402131755672012177131901533_01-139-022.png	0	PMS	M	0.99434
216840111366964013534861372972012332083634742_01-147-145.png	0	PMS	M	0.99196
305197241710047698291397141061639966410_8gvvf6.png	0	PMS	M	0.98656
216840111366964012558082906712009364112417987_00-084-199.png	0	<u>IDC</u>	F	0.98520
216840111366964013575787468332013022090927522_02-013-117.png	0	PMS	M	0.98417
216840111366964013076187734852011292135136135_00-197-078.png	0	PMS	M	0.98359
216840111366964013451228379692012276102128460_01-122-087.png	0	PMS	M	0.98321
216840111366964013589729908692013023185309572_02-012-044.png	0	PMS	M	0.98320
216840111366964013076187734852011297083000255_00-193-002.png	0	PMS	M	0.98309
216840111366964013307756408102012093100450138_01-111-018.png	0	PMS	M	0.98075
216840111366964014008416513202014176103219675_01-192-116.png	0	PMS	M	0.98061
216840111366964013575787468332013017095541406_02-012-130.png	0	PMS	M	0.97715
216840111366964012339356563862009068132653048_00-050-189.png	0	<u>IDC</u>	M	0.97707
216840111366964013575787468332013013111520791_02-009-060.png	0	PMS	M	0.97545
Most uncertain in predicting pneumothorax for PadChest (probability closest to 0.5)				
216840111366964012339356563862009055140056025_00-054-109.png	0	PMS	F	0.50005
216840111366964012768025509942010209091203081_03-141-079.png	0	PMS	M	0.49994
216840111366964012989926673512011104123847985_00-191-127.png	0	PMS	M	0.49993
216840111366964013076187734852011200121554317_00-142-013.png	0	PMS	F	0.49992
216840111366964012734950068292010165091230160_04-012-042.png	0	PMS	F	0.49989
216840111366964013686042548532013260112959079_02-022-171.png	0	<u>IDC</u>	F	0.50021
106517311601416387046892391955760230481_2_pgm35h.png	0	PMS	F	0.50022
216840111366964013962490064942014133091728399_01-168-181.png	0	PMS	F	0.50022
89720507814839131967878326461565402484_hj6b77.png	0	PMS	F	0.50022
216840111366964013686042548532013178095141420_02-097-010.png	0	PMS	M	0.49972
12752243479320242379927_02-085-128.png	0	<u>IDC</u>	M	0.50029
216840111366964013451228379692012256143048163_01-058-017.png	0	PMS	M	0.50031
175106618242165053285074475553992129531_3trl79.png	0	PMS	M	0.50034

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Image Identifier	True Label	Scanner Type	Patient Sex	Probability
216840111366964012734950068292010144084710277_03-143-087.png	0	PMS	F	0.50037
72423424156415717957026223711233354383_ssnu8m.png	0	PMS	F	0.49961
216840111366964012819207061112010299131727211_04-022-145.png	0	IDC	F	0.50047
216840111366964013200840352202011310114532047_01-031-173.png	0	PMS	M	0.49951
216840111366964013829543166512013336101728693_02-093-170.png	0	PMS	F	0.49946
216840111366964012373310883942009084124502690_00-046-124.png	0	PMS	F	0.50056
216840111366964012339356563862009076140153042_00-071-074.png	0	PMS	F	0.49944

C PadChest Projections for Cardiomegaly

Table 8: PadChest distribution across X-ray projections, stratified by cardiomegaly as the target label. Samples cover the full PadChest dataset (after preprocessing).

Projection	No cardiomegaly	Cardiomegaly	Pos. Ratio (%)
PA	81872	8,319	9.2
AP	3,972	333	7.4
AP-horizontal	12,582	1,008	7.7